

NETWORK SECURITY — PRACTICAL LAB REPORT

Cisco Site-to-Site IPsec VPN

Design, Configuration, and Verification of a Site-to-Site IPsec VPN Tunnel Between R1 and R3 Across a Transit Network (R2) — Cisco Packet Tracer

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1. Introduction

A Virtual Private Network (VPN) is a technology that creates a logically private, authenticated communication path across a network that is otherwise untrusted, such as the public Internet or, as in this lab, an intermediate transit network. Rather than leasing dedicated private circuits between sites, a VPN uses encryption and encapsulation to make traffic between two endpoints confidential and tamper-resistant while it physically travels over shared infrastructure.

A Site-to-Site VPN specifically connects two networks — typically a headquarters site and a branch/remote site — through a permanent tunnel established between two gateway devices (in this case, two routers). Unlike a remote-access VPN, individual hosts on either LAN are unaware that a VPN exists; the routers transparently encrypt and decrypt traffic on their behalf. Organizations use Site-to-Site VPNs to interconnect branch offices securely and cost-effectively, avoiding expensive leased lines while still protecting sensitive inter-office traffic (financial data, internal applications, authentication traffic) from interception or modification while it crosses a network the organization does not fully control.

IPsec (Internet Protocol Security) is the framework used in this lab to build that secure tunnel. It operates at the network layer and provides three core security services: confidentiality (encryption of the payload), data integrity and origin authentication (via cryptographic hashing), and peer authentication (proving both routers are who they claim to be before any user data is exchanged). Because IPsec works below the application layer, it protects any IP traffic between the two LANs without requiring changes to the applications or hosts themselves.

Purpose of this Lab

This lab builds and configures a small multi-router network and then implements a Site-to-Site IPsec VPN between two edge routers, R1 and R3, while a third router, R2, acts purely as an intermediate transit hop with no awareness of the VPN. The lab demonstrates the complete lifecycle of an IPsec VPN: basic IP connectivity and dynamic routing (OSPF) as prerequisites, IKE Phase 1 negotiation parameters, IPsec Phase 2 (transform set) parameters, the access list that defines which traffic is actually protected ("interesting traffic"), and the crypto map that ties all of these elements together and binds them to an interface. The lab then verifies the configuration using Cisco IOS show commands and tests the tunnel by generating traffic between the two protected LANs.

This report documents what Team 3 actually configured and observed while completing the lab in Cisco Packet Tracer, distinguishing clearly between configuration that was performed, concepts being explained for context, and results that were verified against the evidence captured in our own CLI screenshots. Where the evidence does not support a claim (for example, where a security association was expected but not observed), this report states that explicitly rather than assuming a successful outcome.

2. Lab Objectives

The lab is organized into two parts, each demonstrating a distinct set of engineering skills:

Part 1 — Basic Network Configuration

- Basic router configuration — hostnames, passwords, and administrative access hardening, establishing a secure management baseline before any VPN work begins.
- Interface addressing — assigning and activating the IP addresses that make up the LAN and WAN segments of the topology.
- OSPF configuration — enabling dynamic routing so that the two protected LANs (192.168.1.0/24 and 192.168.3.0/24) can reach each other across the transit network.
- Basic connectivity verification — confirming that plain, unencrypted IP connectivity exists end-to-end before layering IPsec on top of it.

Part 2 — Site-to-Site IPsec VPN

- IKE/ISAKMP configuration — defining how the two VPN peers authenticate each other and negotiate a secure control channel (Phase 1).
- IPsec configuration — defining the transform set used to encrypt and authenticate the actual data traffic (Phase 2).
- Interesting traffic definition — using an extended ACL to specify exactly which traffic between the two LANs must be protected by the tunnel.
- Crypto map configuration — binding the peer, the interesting-traffic ACL, and the transform set together and applying that policy to the outbound WAN interface.
- VPN verification — using `show crypto isakmp policy`, `show crypto ipsec transform-set`, `show crypto map`, `show crypto isakmp sa`, and `show crypto ipsec sa` to confirm configuration and (attempt to) confirm tunnel state.
- Testing encrypted traffic — generating traffic that matches the ACL and inspecting whether Security Associations (SAs) form and whether packet counters increase.

3. Network Topology

The topology consists of three Cisco ISR 4331 routers and two Cisco Catalyst 2960 switches, connecting two end hosts. PC-A sits on R1's LAN (192.168.1.0/24) behind Switch 1, and PC-C sits on R3's LAN (192.168.3.0/24) behind Switch 3. R1 and R3 are each connected to R2 over a point-to-point WAN transit link (10.1.1.0/30 and 10.2.2.0/30 respectively). R2 has no LAN of its own in this topology and exists solely to route between the two WAN links.

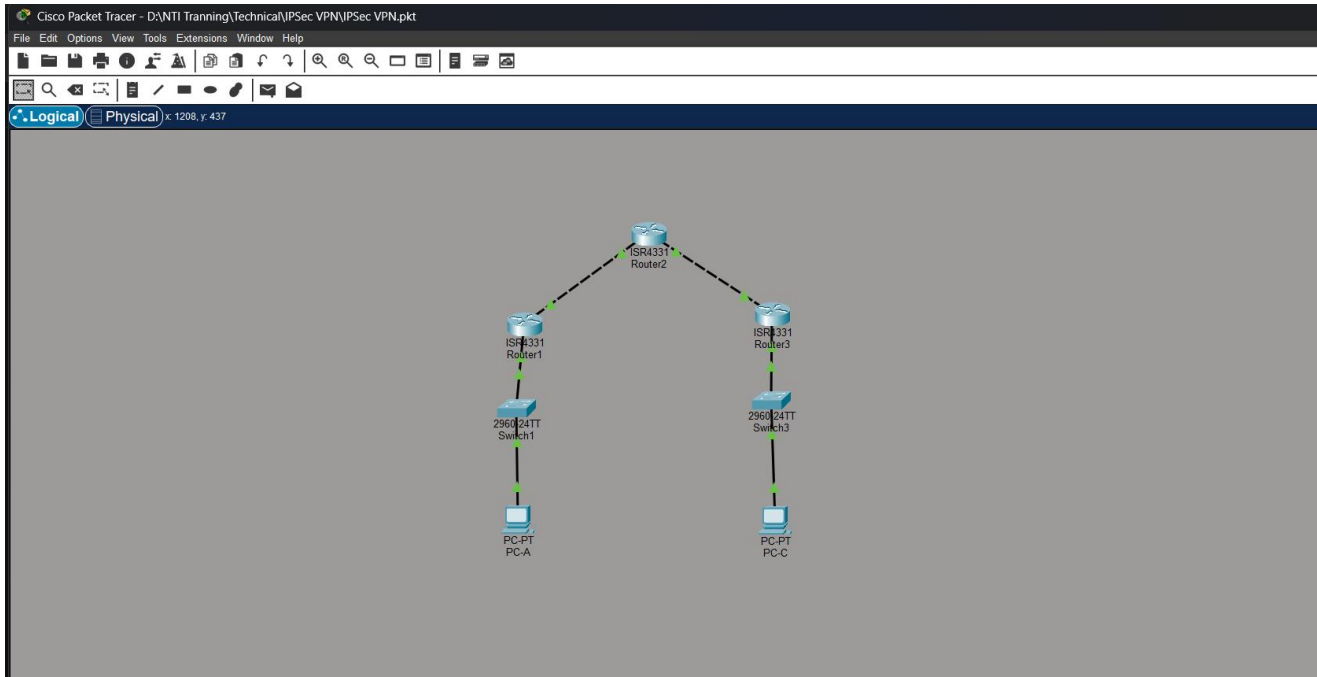


Figure 1 — Packet Tracer logical topology: PC-A — Switch 1 — R1 — R2 — R3 — Switch 3 — PC-C. R1 and R3 are the IPsec VPN peers; R2 is a pass-through transit router with no IPsec configuration.

The end-to-end traffic path for a packet sent from PC-A to PC-C is:

PC-A -> Switch 1 -> R1 (G0/0/1 in / G0/0/0 out) -> R2 -> R3 (G0/0/0 in / G0/0/1 out) -> Switch 3 -> PC-C

Two points are important to understand about this design:

- R1 and R3 are the IPsec VPN peers/endpoints. The crypto map that builds the tunnel is applied only to R1's G0/0/0 interface and R3's G0/0/0 interface — specifically, the WAN-facing interface each router uses to reach the other peer's WAN address (10.1.1.1 and 10.2.2.1).
- R2 is only a transit/pass-through router. It forwards IP packets between the two WAN links using OSPF-learned routes and has no crypto configuration at all — it is completely unaware that the traffic it forwards is (or is intended to be) encrypted.
- The protected traffic is defined as anything sourced from 192.168.1.0/24 and destined to 192.168.3.0/24, and vice versa — i.e., LAN-to-LAN traffic between PC-A's and PC-C's subnets.
- The IPsec tunnel is a logical construct between R1 (10.1.1.1) and R3 (10.2.2.1); it does not appear as a separate link in the topology diagram — it rides on top of the existing R1–R2–R3 physical path.

4. Addressing Table

The addressing plan below is taken directly from the lab's official Addressing Table and matches the interface addressing confirmed on our own routers via show ip interface brief (Figures 2–4).

Device	Interface	IP Address	Subnet Mask	Default GW	Switch Port	Role
R1	G0/0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/5	LAN gateway
R1	G0/0/0	10.1.1.1	255.255.255.252	N/A	N/A	VPN / WAN endpoint
R2	G0/0/0	10.1.1.2	255.255.255.252	N/A	N/A	Transit (to R1)
R2	G0/0/1	10.2.2.2	255.255.255.252	N/A	N/A	Transit (to R3)
R3	G0/0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5	LAN gateway
R3	G0/0/0	10.2.2.1	255.255.255.252	N/A	N/A	VPN / WAN endpoint
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/6	Protected host (LAN-A)
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18	Protected host (LAN-C)

Reading the Addressing Plan

- LAN addressing: 192.168.1.0/24 (R1 side, hosting PC-A) and 192.168.3.0/24 (R3 side, hosting PC-C) are the two networks the VPN is designed to protect.
- WAN / transit addressing: 10.1.1.0/30 (R1–R2) and 10.2.2.0/30 (R2–R3) are /30 point-to-point links used purely for inter-router routing; no VPN protection is applied to traffic exchanged on these links directly (e.g., router-to-router pings), only to LAN-to-LAN traffic that is routed across them.
- PC default gateways: PC-A points to 192.168.1.1 (R1) and PC-C points to 192.168.3.1 (R3) — each host's local router.
- VPN endpoints: the IPsec peer addresses are R1's G0/0/0 (10.1.1.1) and R3's G0/0/0 (10.2.2.1) — the addresses each router uses to reach the other across the transit network, and the addresses referenced in the crypto isakmp key and set peer commands.

5. Part 1 — Basic Device Configuration

Before any VPN configuration is attempted, each router (R1, R2, R3) was given a basic configuration baseline: hostnames matching the topology, interface IP addressing per the table above, DNS-lookup suppression, and local administrative security.

```
hostname R1
...
no ip domain lookup
```

Disabling DNS lookup with no ip domain lookup prevents the router from attempting to resolve a mistyped command as a hostname and pausing for a DNS timeout — a small but important quality-of-life step when configuring routers interactively over the CLI, since a typo would otherwise stall the terminal for tens of seconds.

Interface addressing was applied per device, e.g. on R1:

```
interface GigabitEthernet0/0/1
 ip address 192.168.1.1 255.255.255.0
 no shutdown
interface GigabitEthernet0/0/0
 ip address 10.1.1.1 255.255.255.252
 no shutdown
```

Local administrative security was also configured on R1 and R3, consistent with the lab specification: a minimum enforced password length, an enable secret hashed with the type 9 (SCRYPT) algorithm, and a local user account (admin01) also hashed with type 9, used for console and SSH login. The console line was set to use the local user database with a five-minute inactivity timeout and logging synchronous to keep unsolicited log messages from interrupting command entry. SSH was enabled with a 2048-bit RSA key pair, restricted to SSHv2, using the domain name netsec.com.

Evidence: the login sequence captured on both R1 and R3 (Figures 2–3) explicitly prompts for Username: admin01 followed by a Password: prompt before reaching the R1# / R3# privileged-mode prompt after en. This confirms that local AAA login and an enable secret were both successfully applied and functioning, even though the underlying running-config for these lines was not itself captured in a separate screenshot in this submission.

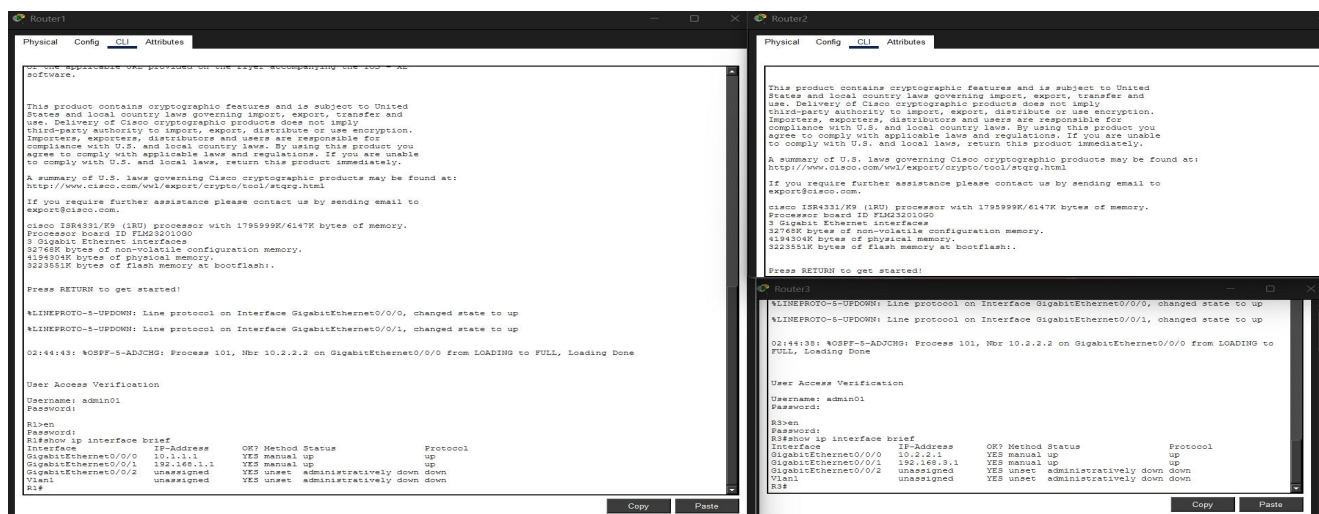


Figure 2 — R1 and R2 initial boot sequence and login. Interfaces on R1 show G0/0/0 (10.1.1.1) and G0/0/1 (192.168.1.1) up/up, and successful authentication with the local admin01 account is visible.

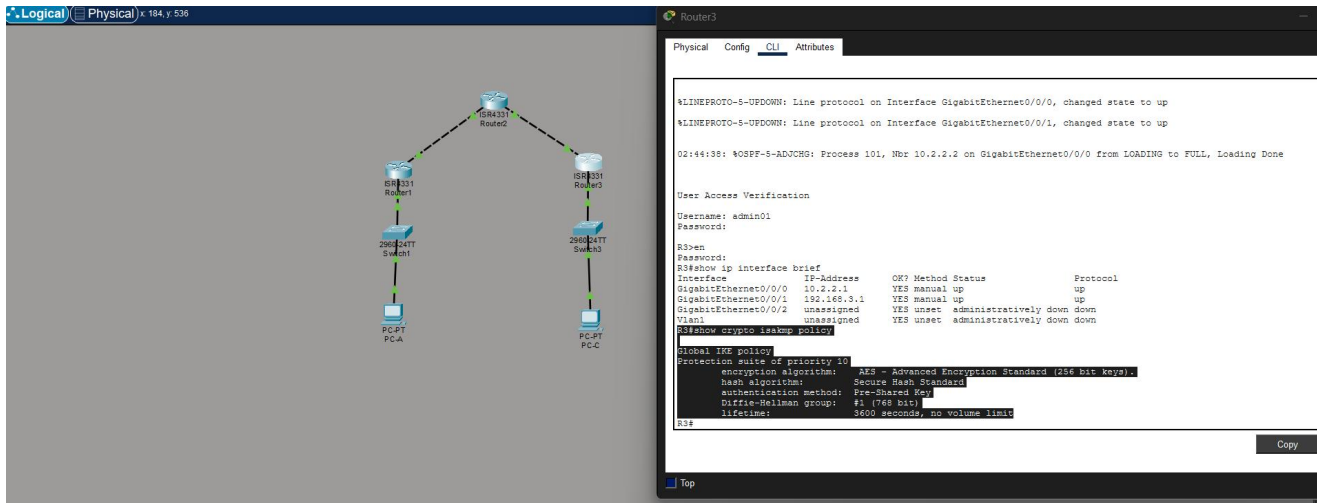


Figure 3 — R3 boot sequence, OSPF adjacency forming, local login (admin01), and show ip interface brief confirming G0/0/0 (10.2.2.1) and G0/0/1 (192.168.3.1) up/up.

6. OSPF Routing Configuration and Verification

OSPF is configured before the VPN for a simple reason: IPsec protects traffic, it does not route it. Without a working routing protocol giving R1 a path to 192.168.3.0/24 (and R3 a path to 192.168.1.0/24), there would be no IP reachability between the two LANs for the crypto map to protect in the first place. OSPF process 101 in area 0 was enabled on all three routers:

```
! R1
router ospf 101
 network 192.168.1.0 0.0.0.255 area 0
 network 10.1.1.0 0.0.0.3 area 0

! R2
router ospf 101
 network 10.1.1.0 0.0.0.3 area 0
 network 10.2.2.0 0.0.0.3 area 0

! R3
router ospf 101
 network 192.168.3.0 0.0.0.255 area 0
 network 10.2.2.0 0.0.0.3 area 0
```

R2's role here is purely that of a Layer 3 forwarder: it advertises both of its transit links into OSPF area 0 and therefore learns, and propagates, routes to both LANs, without ever having a route to advertise for a LAN of its own.

Verification

On R1, show ip ospf neighbor confirms a FULL adjacency with R2 (10.2.2.2, seen as neighbor ID 10.2.2.2 in FULL/BDR state on GigabitEthernet0/0/0), and show ip route shows the OSPF-learned route to

192.168.3.0/24 via 10.1.1.2 (R2) with the expected O code and administrative distance/metric [110/3]. This single route entry is the proof that inter-LAN reachability exists at the IP layer, independent of the VPN.

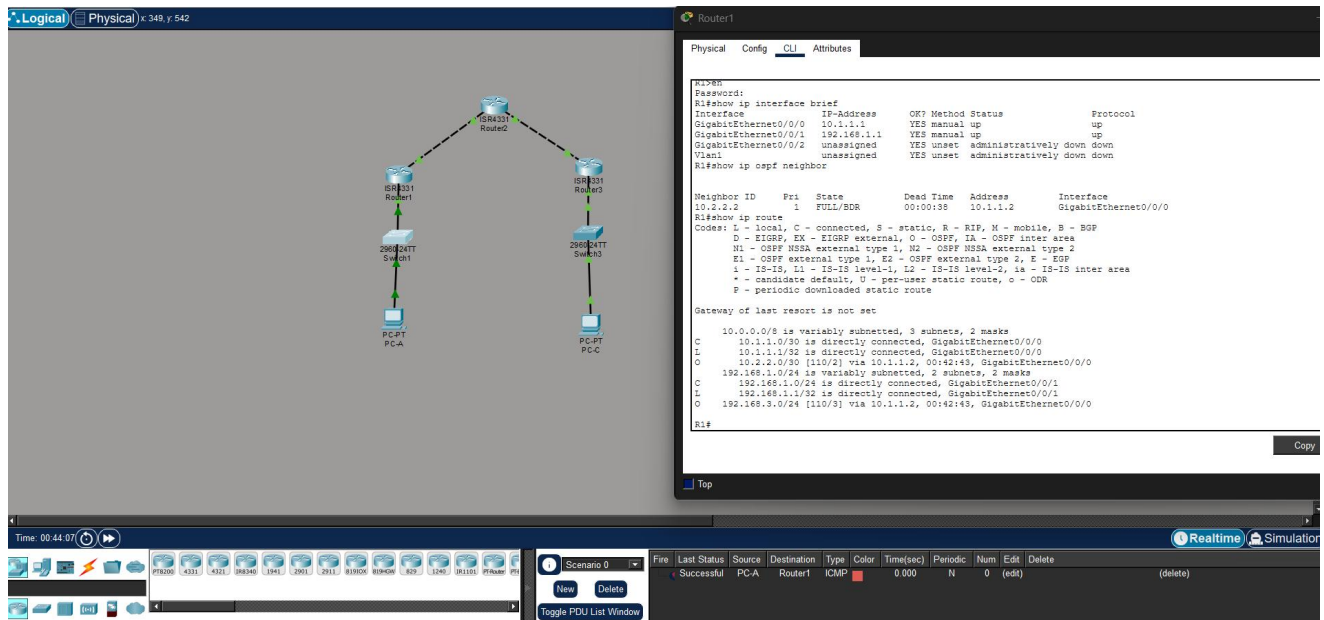


Figure 4 — R1: show ip interface brief, show ip ospf neighbor (FULL adjacency with 10.2.2.2), and show ip route, showing the OSPF-learned route to 192.168.3.0/24 via 10.1.1.2. The PDU list confirms a successful ICMP echo from PC-A to R1.

7. Basic Connectivity Verification

Before enabling any IPsec configuration, the lab requires confirming plain connectivity between PC-A and PC-C, since a working VPN can only be built on top of working, unencrypted IP connectivity. Figure 4 (above) already shows a successful PC-A → R1 ping. Full end-to-end connectivity from PC-A across the entire path (through R1, R2, and R3, to PC-C) was subsequently confirmed with a batch of simple PDUs launched from PC-A:

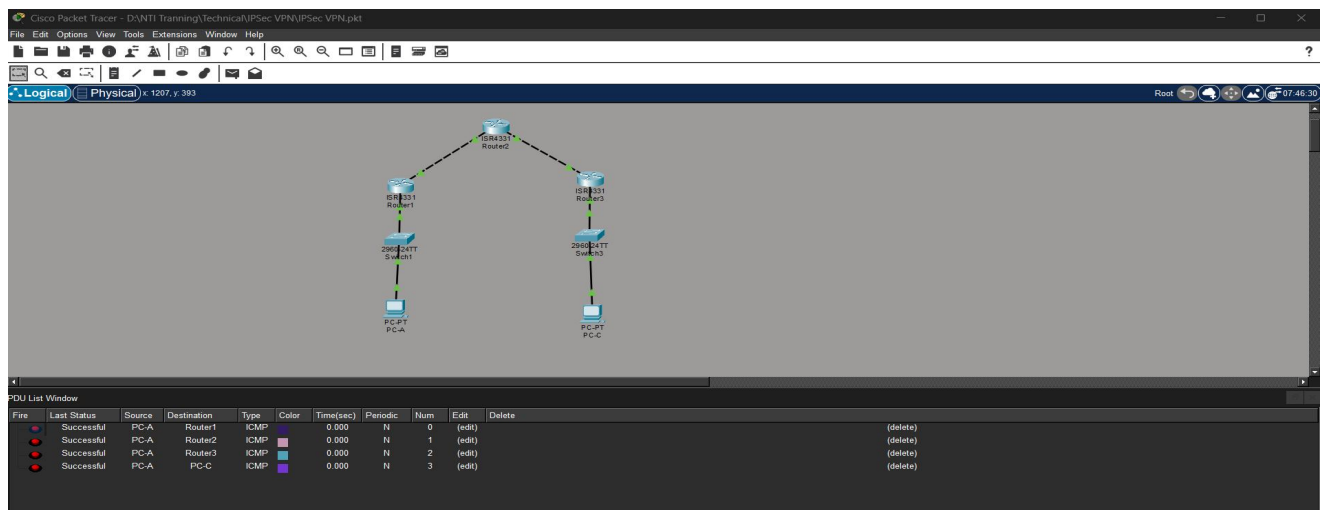


Figure 5 — Packet Tracer PDU List Window: four simple ICMP PDUs sent from PC-A — to R1, R2, R3, and finally PC-C — all reporting Successful. This confirms baseline IP reachability across the full topology, including the final PC-A → PC-C hop.

This result demonstrates that OSPF is functioning correctly end-to-end and that, at this stage, the network can already carry traffic between the two LANs — necessarily still unencrypted, since the crypto map had not yet been fully activated with matching Security Associations at this point in the configuration. This is the expected and required baseline before layering IPsec on top.

8. IPsec VPN Architecture Overview

Before looking at individual commands, it helps to understand the two-stage design that every Cisco IOS site-to-site IPsec deployment follows.

IKE Phase 1 — Building the Control Channel

Internet Key Exchange (IKE) Phase 1 establishes a secure, authenticated management channel between the two VPN peers, R1 and R3. During this phase, the two routers negotiate and agree on an ISAKMP policy — the encryption algorithm, hash algorithm, authentication method, and Diffie-Hellman group used to protect the negotiation itself and to derive shared keying material. Successful completion of Phase 1 produces an ISAKMP Security Association, but no user data has been protected yet — this phase only builds the secure channel used to negotiate everything else.

IPsec Phase 2 — Protecting Data Traffic

IPsec Phase 2 runs inside the Phase 1 channel and negotiates the actual parameters used to protect user (data-plane) traffic: the transform set (which encryption/authentication algorithms are applied to each packet), and — via the crypto map — which specific traffic qualifies for protection (the interesting-traffic ACL) and what peer and lifetime apply. Successful Phase 2 negotiation produces a pair of unidirectional IPsec Security Associations (inbound and outbound) that are then used to encrypt and decrypt matching packets.

In this lab, Phase 1 is configured with crypto isakmp policy, Phase 2 is configured with crypto ipsec transform-set, and the two are tied together — along with the peer address and the interesting-traffic ACL — by the crypto map, which is then applied to the physical interface facing the remote peer.

9. IKE / ISAKMP Phase 1 Configuration

An ISAKMP policy was created under policy number 10 on both R1 and R3. Each parameter of the policy governs a different security property of the Phase 1 negotiation:

- Policy priority (10) — the number Cisco IOS uses to order multiple candidate policies; peers negotiate down the list from lowest number (highest priority) until a match is found.
- authentication pre-share — peers prove their identity to each other using a shared secret (pre-shared key) rather than digital certificates.
- encryption — the cipher protecting the ISAKMP negotiation itself.
- hash — the algorithm providing integrity checking of the ISAKMP messages.
- group — the Diffie-Hellman group used to derive the shared secret without transmitting it directly across the network; a higher group number generally provides a mathematically stronger key at the cost of more CPU.

- lifetime — how long (in seconds) the resulting ISAKMP SA remains valid before it must be renegotiated.

The lab specification calls for hash sha, authentication pre-share, group 24, lifetime 3600, and encryption aes 256. This is the configuration intent that was applied on both R1 and R3:

```
crypto isakmp policy 10
 hash sha
 authentication pre-share
 group 24
 lifetime 3600
 encryption aes 256
```

Verification — show crypto isakmp policy

R3's actual verification output was captured directly from the CLI and is shown in Figure 3 (Section 5) and again below alongside R1's matching output. It confirms AES-256 encryption, SHA hashing, pre-shared key authentication, and a 3600-second lifetime, exactly as specified — with one discrepancy worth flagging: the Diffie-Hellman group reported is #1 (768-bit), not group 24 (2048-bit) as instructed in the lab. Because this value is identical on both R1 and R3 in our captured evidence, Phase 1 negotiation is not blocked by a mismatch between the peers — but it means the actual DH strength implemented in this run is weaker than the lab's target configuration. This is recorded here rather than silently corrected, consistent with reporting only what the evidence shows.

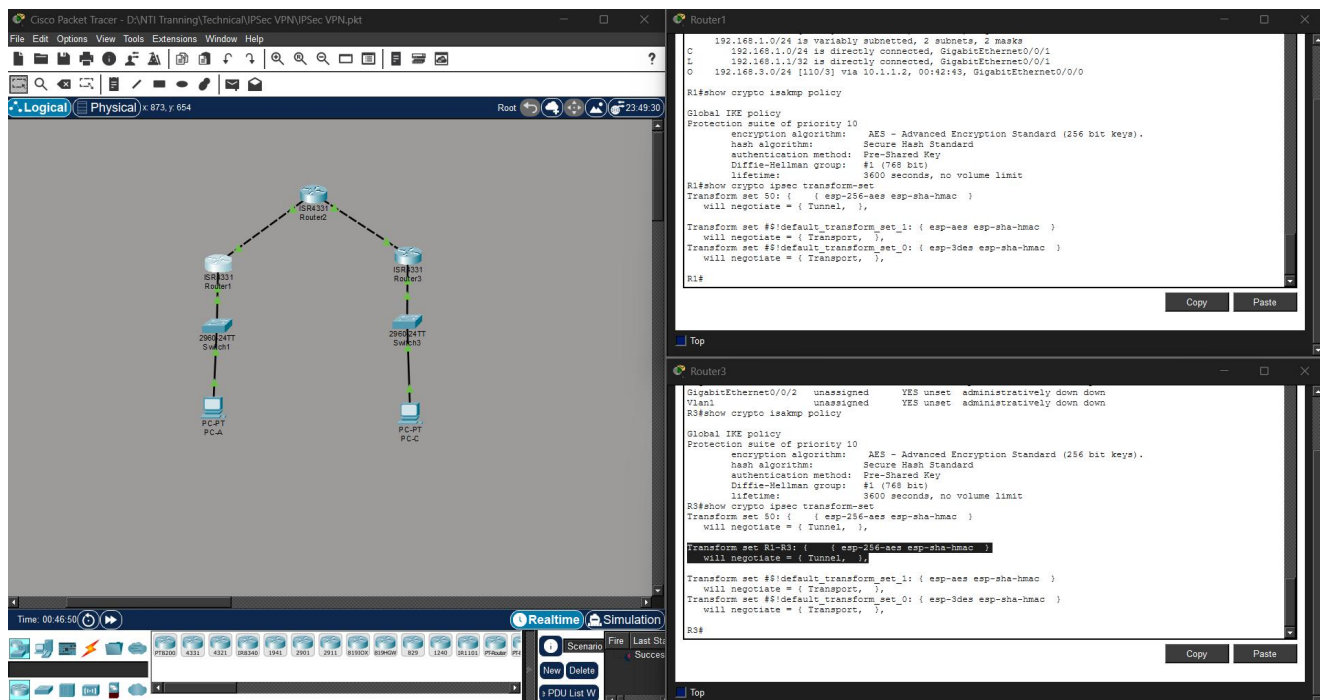


Figure 6 — Side-by-side show crypto isakmp policy and show crypto ipsec transform-set on R1 (top) and R3 (bottom). Both peers report identical Phase 1 parameters: AES-256 encryption, SHA hashing, pre-shared key authentication, DH group #1 (768-bit), 3600-second lifetime.

Q: Which IP addresses should be used to configure the IKE peers, given the topology and addressing table?

A: The IP address of the remote peer's WAN-facing (crypto-map) interface — the address the local router actually

uses to route traffic toward the other end. From R1's perspective that is R3's G0/0/0 address, 10.2.2.1; from R3's perspective it is R1's G0/0/0 address, 10.1.1.1. These are the same addresses referenced later in the `crypto isakmp key` and `set peer` commands.

10. Pre-Shared Key Configuration

Because the ISAKMP policy specifies pre-share as the authentication method, each router must be told what secret key to expect from its peer, and which peer address that key applies to. The pre-shared key must match exactly, character for character, on both ends — any mismatch causes Phase 1 authentication to fail even if every other policy parameter matches perfectly.

```
! On R1 — points to R3's WAN address
crypto isakmp key [REDACTED] address 10.2.2.1

! On R3 — points to R1's WAN address
crypto isakmp key [REDACTED] address 10.1.1.1
```

The key string itself is masked as [REDACTED] in this report rather than reproduced, even though the lab's reference value is a simple training key; the important architectural point is that the key is address-bound (tied specifically to the remote peer's IP) and must be identical on both routers, not the literal string used.

11. IPsec Transform Set (Phase 2)

The transform set defines the combination of protocols and algorithms applied to actual user data once the tunnel is up. This lab uses ESP (Encapsulating Security Payload), which — unlike AH (Authentication Header) — provides both confidentiality (encryption) and integrity/authentication in a single protocol, making it the standard choice for IPsec VPNs that need to keep payload contents private.

```
crypto ipsec transform-set R1-R3 esp-256-aes esp-sha-hmac
```

- `esp-256-aes` — encrypts the packet payload using AES with a 256-bit key, providing confidentiality.
- `esp-sha-hmac` — computes an HMAC-SHA hash over each packet to detect any tampering or corruption in transit, providing integrity and data-origin authentication.

As with the ISAKMP policy, the transform set's algorithm parameters must match on both peers, or Phase 2 negotiation will fail. Our captured evidence (Figure 6, above) shows this was achieved, though with a naming inconsistency worth noting for completeness: R1's transform set is tagged 50, while R3 carries both a transform set named 50 and a second, separately-configured transform set named R1-R3 — both instances resolving to the same underlying parameters, { `esp-256-aes esp-sha-hmac` }, negotiating in Tunnel mode. Because a transform-set tag is only locally significant (Cisco IOS peers negotiate on the algorithm content of the transform set, not its name), this naming mismatch does not by itself prevent Phase 2 negotiation, but it

does deviate from the lab's instruction that both routers use the identical tag R1-R3, and reflects a configuration performed slightly differently than instructed.

The IPsec security-association lifetime (separate from the ISAKMP lifetime in Section 9) was also set to 1800 seconds (30 minutes) per the lab specification:

```
crypto ipsec security-association lifetime seconds 1800
```

Q: What is the function of the IPsec transform set?

A: It defines the exact combination of security protocol (AH and/or ESP), encryption cipher, and hashing/authentication algorithm used to protect data-plane traffic once the tunnel is established. Both peers must offer at least one matching transform set during Phase 2 negotiation, or no IPsec Security Association can be built.

12. Interesting Traffic — Crypto Access List

Not all traffic that crosses R1's or R3's WAN interface should be encrypted — only traffic between the two protected LANs. "Interesting traffic" is the term Cisco IOS uses for any packet that matches a permit statement in the ACL referenced by the crypto map; only that traffic triggers IKE negotiation and is subsequently encrypted. Traffic that does not match the ACL (or that matches an explicit deny) is simply forwarded normally, in plaintext — it is not dropped.

```
! On R1
access-list 101 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255

! On R3 — mirrored
access-list 101 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

The two ACLs are deliberately mirror images of each other: R1's ACL matches traffic from its own LAN toward R3's LAN, while R3's ACL matches the reverse direction. This mirroring is required because each router evaluates "interesting traffic" from its own local point of view — for the tunnel to work symmetrically in both directions, both routers must agree on the same conversation, just described from opposite ends.

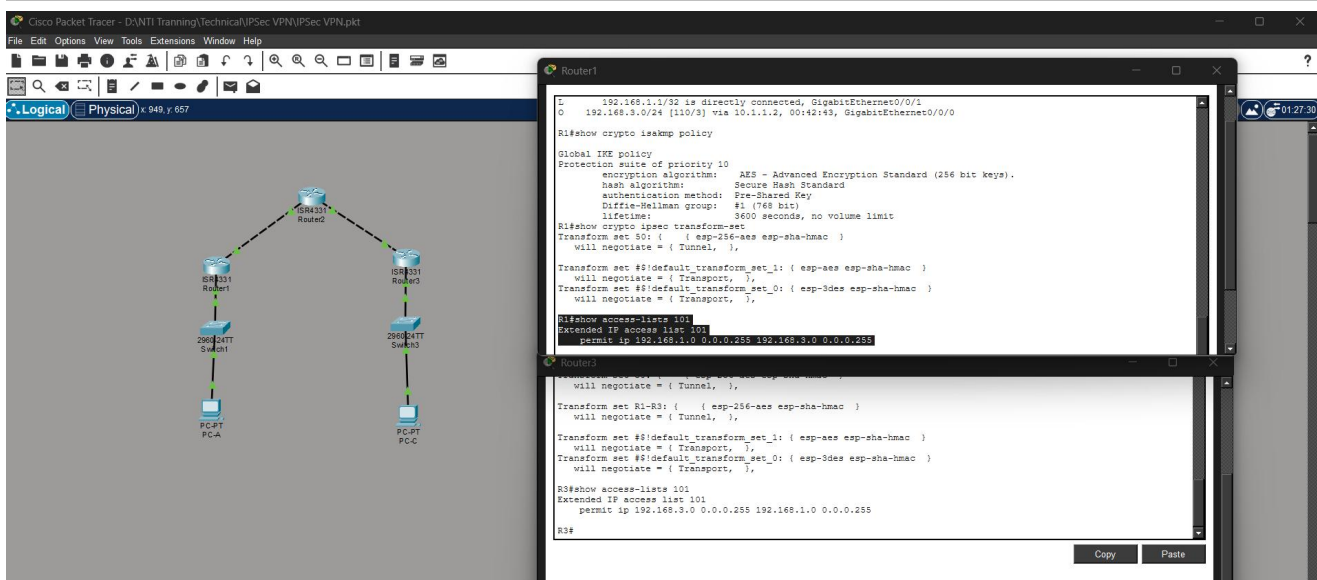


Figure 7 — show access-lists 101 captured on R1 (top, permitting 192.168.1.0/24 → 192.168.3.0/24) and R3 (bottom, permitting 192.168.3.0/24 → 192.168.1.0/24), confirming the mirrored ACL configuration required for interesting traffic to be recognized symmetrically.

Q: Does IPsec evaluate whether the access lists are mirrored, as a requirement to negotiate its security association?

A: Yes — Cisco IOS expects the crypto ACLs on the two peers to be symmetric (mirror images) of one another. If the ACLs do not describe the same conversation from each router's own perspective, the two ends will disagree about what traffic the SA should protect, and Phase 2 negotiation for that traffic selector will fail or the resulting SA will not match the traffic actually being sent.

13. Crypto Map Configuration

The crypto map is the object that ties every previous piece of configuration together: which remote peer to negotiate with, which traffic to protect (the ACL), which transform set to use, whether Perfect Forward Secrecy (PFS) is required, and what SA lifetime to apply. It is then bound to a physical interface, which tells IOS to evaluate this policy against traffic leaving (or destined toward) that interface.

```
! R1
crypto map CMAP 10 ipsec-isakmp
 match address 101
 set peer 10.2.2.1
 set pfs group24
 set transform-set R1-R3
 set security-association lifetime seconds 900
interface GigabitEthernet0/0/0
 crypto map CMAP

! R3 — mirrored
crypto map CMAP 10 ipsec-isakmp
 match address 101
 set peer 10.1.1.1
 set pfs group24
 set transform-set R1-R3
 set security-association lifetime seconds 900
```

```
interface GigabitEthernet0/0/0
crypto map CMAP
```

- match address 101 — links the crypto map to the interesting-traffic ACL from Section 12.
- set peer — identifies the remote router's WAN address that this policy negotiates with.
- set pfs group24 — enables Perfect Forward Secrecy using DH group 24 for Phase 2, meaning a fresh Diffie-Hellman exchange (independent of the Phase 1 key) is performed for every new IPsec SA, so that compromise of one session's key does not expose past or future sessions.
- set transform-set — selects which Phase 2 transform set (Section 11) this map uses.
- set security-association lifetime seconds 900 — overrides the global 1800-second IPsec SA lifetime with a shorter, map-specific value of 15 minutes.

The crypto map is applied to the interface that faces the remote peer across the transit network — R1's G0/0/0 and R3's G0/0/0 — because that is the interface each router uses to route toward the peer's WAN address; applying it to the LAN-facing interface would have no effect on traffic destined for the remote site. (The lab manual's step-by-step instructions list R3's crypto map interface as G0/0/1, but this appears to be an error in the source document: both the addressing table and R3's own show crypto map output — Interfaces using crypto map CMAP: GigabitEthernet0/0/0 — confirm G0/0/0 is the correct, WAN-facing interface actually used.)

Verification — show crypto map

R1's captured output confirms every element of the intended policy: peer 10.2.2.1, ACL 101 with the correct permit statement, a security-association lifetime of 4,608,000 kilobytes / 900 seconds, PFS enabled (Y) using DH group24, the R1-R3-equivalent transform set (tagged 50 in this router's configuration — see Section 11), and the map correctly applied to GigabitEthernet0/0/0.

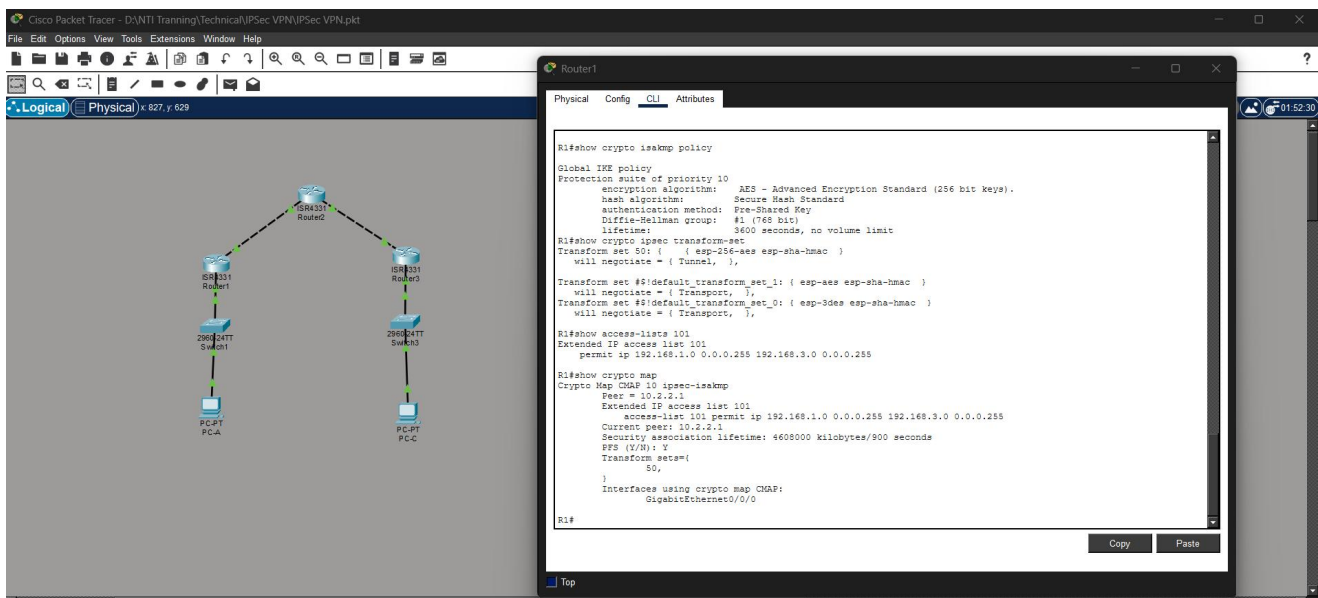


Figure 8 — R1: consolidated verification of `show crypto isakmp policy`, `show crypto ipsec transform-set`, `show access-lists 101`, and `show crypto map`. The crypto map output confirms Peer = 10.2.2.1, ACL 101, PFS = Y (group24), and the map bound to GigabitEthernet0/0/0.

14. VPN Verification — Before Interesting Traffic

Applying a crypto map to an interface does not, by itself, establish a VPN tunnel. IOS does not attempt to negotiate IKE or build any Security Associations until it actually sees a packet that matches the interesting-traffic ACL. Immediately after configuration and before any qualifying traffic was generated, show crypto isakmp sa and show crypto ipsec sa were issued on R1 to capture this "before" state.

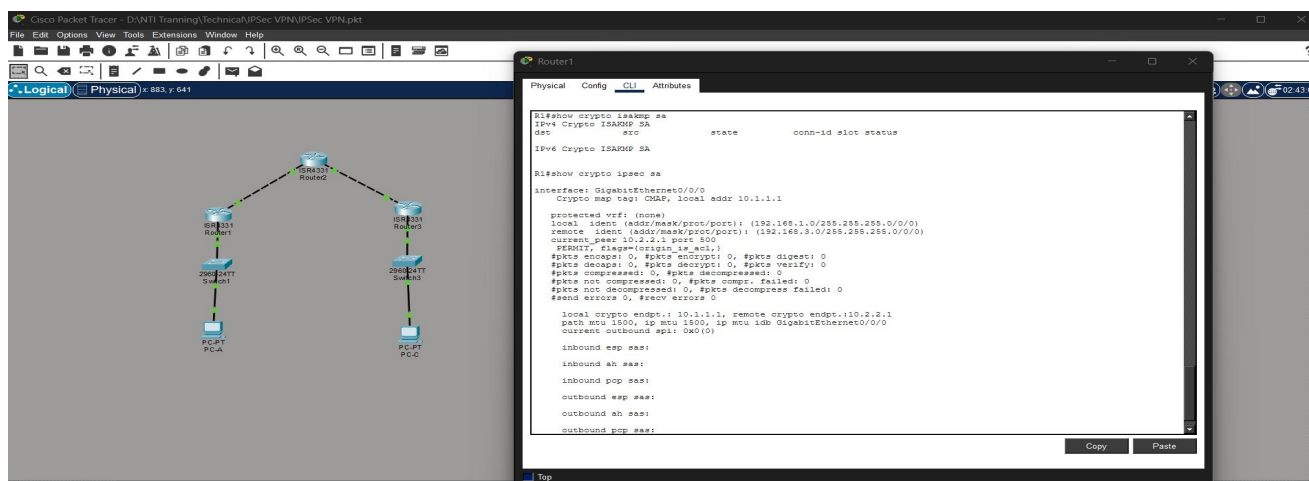


Figure 9 — R1 before interesting traffic: show crypto isakmp sa returns an empty IKE SA table, and show crypto ipsec sa shows the crypto map bound correctly to GigabitEthernet0/0/0 with the correct local/remote identities (192.168.1.0/24 ↔ 192.168.3.0/24) and peer (10.2.2.1), but all packet counters (#pkts encaps, encrypt, decaps, decrypt) at zero and current outbound spi: 0x0(0).

Q: Why haven't any Security Associations been negotiated at this point?

A: Because Cisco IOS builds IKE and IPsec SAs on demand, triggered only by traffic that matches the crypto map's ACL. Simply configuring and applying the crypto map arms the policy but does not itself initiate negotiation — the tunnel remains dormant until interesting traffic is sent.

15. Generating Interesting Traffic

To trigger negotiation, traffic must be sourced from one protected LAN and destined for the other — i.e., it must match access-list 101. Two categories of traffic were exercised in this lab for comparison:

Uninteresting Traffic (Control Test)

Pings sent directly to a router's own WAN or LAN interface address (rather than through it, LAN-to-LAN) do not match the 192.168.1.0/24 ↔ 192.168.3.0/24 selector in ACL 101, and OSPF hello packets exchanged between routers likewise fall outside that selector. Neither type of traffic is expected to trigger IKE negotiation, regardless of whether the ping itself succeeds at the IP layer.

Q: Was a Security Association created for pings to router interfaces, or for OSPF hello traffic between R1 and R3?

A: No. Neither type of traffic matches the source/destination ranges defined in access-list 101 (192.168.1.0/24 to 192.168.3.0/24). Only traffic actually flowing between the two protected LAN subnets qualifies as interesting traffic; router-to-router control traffic and pings to interface addresses are forwarded normally, without ever consulting the crypto map.

Interesting Traffic — PC-A to PC-C

A batch of ICMP echo requests was sent from PC-A (192.168.1.3) to PC-C (192.168.3.3), which is precisely the traffic pattern ACL 101 is written to match — a source in 192.168.1.0/24 destined for 192.168.3.0/24. This is the same PDU List Window result already presented in Figure 5 (Section 7): all four PDUs, including PC-A → PC-C, are reported Successful, confirming the packets themselves reached their destination.

Q: What other types of traffic would result in an SA forming and tunnel establishment?

A: Any IP traffic sourced from 192.168.1.0/24 and destined for 192.168.3.0/24 (or the reverse), regardless of application — for example, a Telnet or SSH session between the two LANs, an HTTP/HTTPS request, or an FTP transfer — would equally match access-list 101 and qualify as interesting traffic capable of triggering IKE negotiation.

16. IKE and IPsec Security Association Verification

After generating the PC-A → PC-C traffic shown in Figure 5, show crypto isakmp sa and show crypto ipsec sa were re-issued on R1 to check whether an IKE Security Association had formed and whether the IPsec packet counters had incremented.

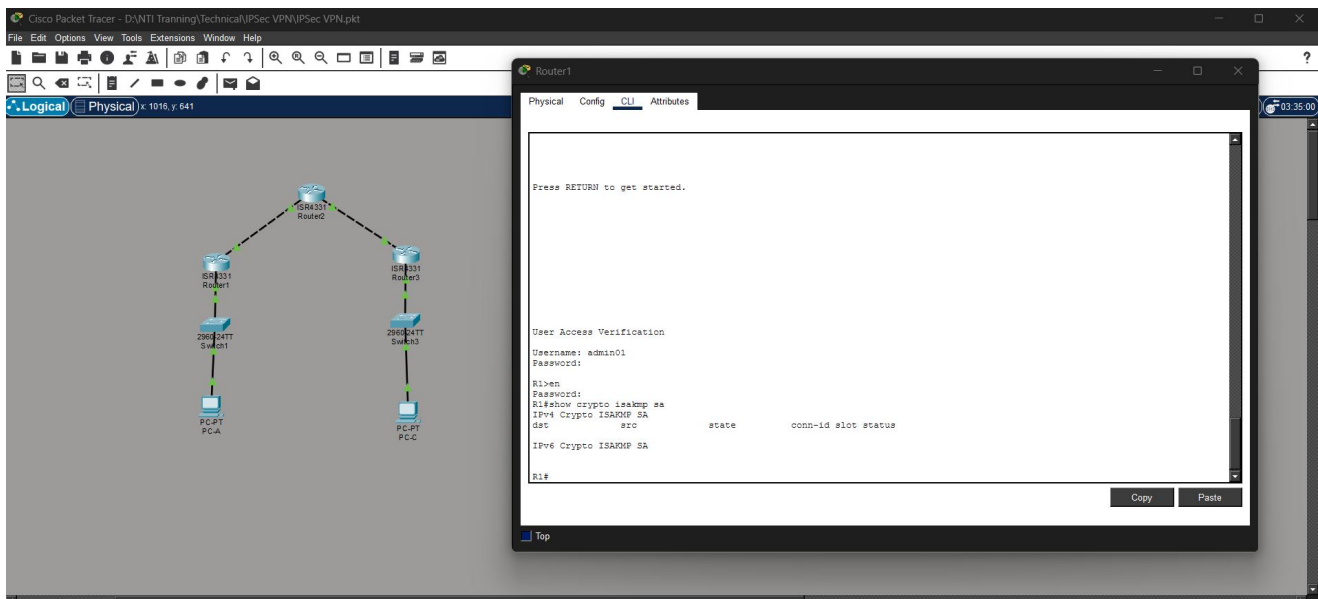


Figure 10 — R1, re-checked after traffic generation: show crypto isakmp sa still returns an empty IKE SA table (no dst / src / state / conn-id entries).

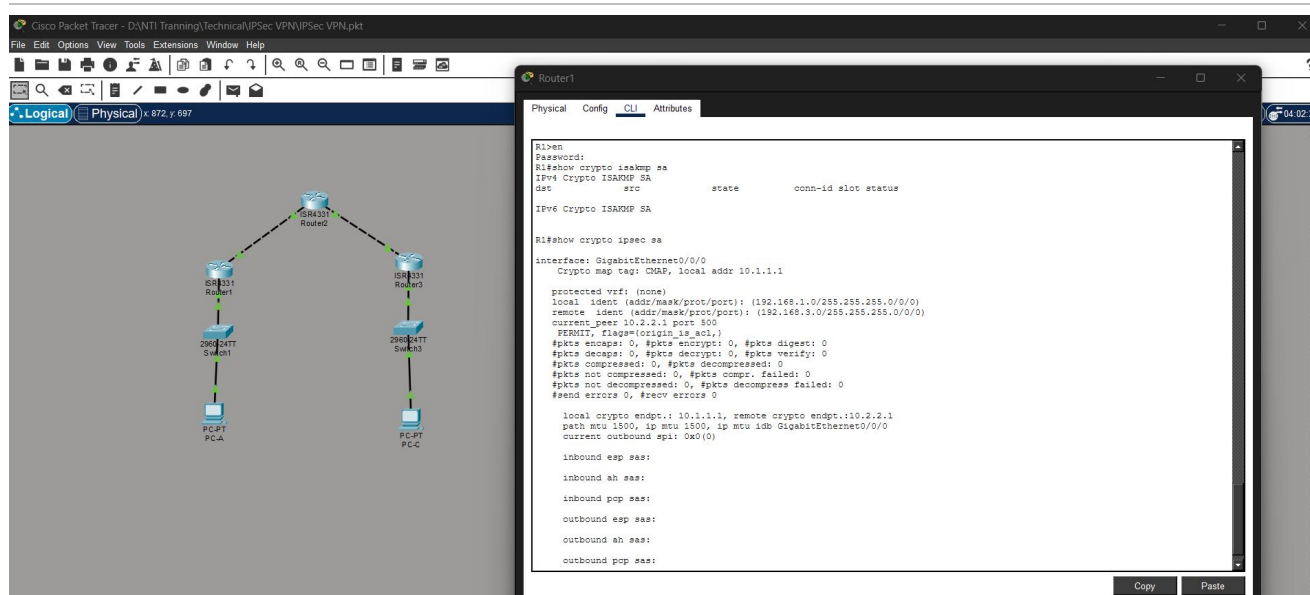


Figure 11 — R1, final verification pass: show crypto isakmp sa remains empty, and show crypto ipsec sa still reports #pkts encaps: 0, #pkts encrypt: 0, #pkts decaps: 0, #pkts decrypt: 0, with current outbound spi: 0x0(0) — unchanged from the pre-traffic baseline in Figure 9.

This is an important, and honest, result to report: based on the evidence captured in this implementation, no ISAKMP SA and no active IPsec SA were established, and the IPsec packet counters did not increase, even though the PC-A → PC-C ICMP traffic itself completed successfully at the IP layer (Figure 5). In a fully working implementation of this lab, the expected output at this stage — per the Cisco reference lab — is a state of QM_IDLE / ACTIVE in show crypto isakmp sa and non-zero #pkts encaps / #pkts decaps counters with a populated inbound/outbound esp sas block in show crypto ipsec sa. That expected reference behavior is explained below for completeness, but it should not be read as a claim about what our own routers actually produced, since our own screenshots do not show it.

Expected Reference Behavior (Not Observed in Our Evidence)

For context, and drawing only on the Cisco lab's documented reference output rather than our own capture, a successfully negotiated tunnel would show a QM_IDLE / ACTIVE entry in show crypto isakmp sa between 10.2.2.1 (dst) and 10.1.1.1 (src), and show crypto ipsec sa would report a non-zero, incrementing pair of encaps/decaps counters together with populated inbound and outbound esp sas blocks listing a negotiated SPI, the negotiated transform (esp-256-aes esp-sha-hmac), Tunnel mode, and Status: ACTIVE(ACTIVE).

Q: Based on the evidence captured, how many packets were transformed (encrypted/decrypted) between R1 and R3?

A: Zero. Both the pre-traffic capture (Figure 9) and the post-traffic capture (Figure 11) show identical #pkts encaps: 0 / #pkts decrypt: 0 counters and an unassigned outbound SPI (0x0(0)). No non-zero counters were captured at any point in this implementation's evidence, so a successful encryption/decryption cycle cannot be confirmed for this run.

17. End-to-End Testing (PC-A ↔ PC-C)

The final connectivity test, shown in Figure 5, confirms that PC-A (192.168.1.3) can successfully reach PC-C (192.168.3.3) with all four ICMP echo requests reported Successful in the PDU List Window. This proves IP-layer, end-to-end connectivity across the entire topology (PC-A → S1 → R1 → R2 → R3 → S3 → PC-C).

It is important to separate two distinct claims that this kind of test can support, and to only make the one the evidence backs:

- **Connectivity:** a successful ping proves that IP packets are being routed correctly end-to-end. Our evidence fully supports this claim (Figure 5).
- **Encryption:** a successful ping does not, by itself, prove that the traffic was actually encrypted by IPsec while crossing the transit network — that requires independent evidence from `show crypto isakmp sa` / `show crypto ipsec sa` showing an active SA and non-zero packet counters. Our evidence (Section 16, Figures 9–11) does not support this second claim for this implementation.

In other words, PC-A successfully reached PC-C, but based on the SA and counter evidence gathered, that traffic cannot be confirmed to have traveled through an active IPsec tunnel in this specific run — it may equally have been forwarded in plaintext by the underlying OSPF-derived route while the crypto map failed to intercept and encrypt it.

18. Before vs. After Analysis

The table below summarizes only what our own screenshots actually show at each stage.

Verification	Before Interesting Traffic	After Interesting Traffic
IKE SA (<code>show crypto isakmp sa</code>)	Empty — no entries	Still empty — no entries (Figures 10–11)
IPsec SA (<code>show crypto ipsec sa</code>)	No active SA; outbound spi 0x0(0)	Unchanged; outbound spi still 0x0(0)
#pkts encaps / encrypt	0	0
#pkts decaps / decrypt	0	0
PC-A → PC-C ping	Not yet tested	Successful (IP-layer connectivity only, Figure 5)

The configuration itself progressed correctly through every stage the lab requires — ISAKMP policy, transform set, ACL, and crypto map were all configured and independently verified as syntactically and logically correct (Sections 9–13). What did not progress, based on this evidence, is the final step: Configured → Triggered → Established → Encrypted Traffic. The chain stops at Triggered — matching traffic was generated (Figure 5) — without visible evidence of Established or Encrypted.

19. Security Analysis

Assuming the configuration operates as designed (i.e., focusing on what was configured rather than what was empirically confirmed to be active), the implemented VPN design provides the following security properties:

- Confidentiality — ESP with AES-256 encryption (esp-256-aes) is configured to protect the payload of traffic between the two LANs from being read if intercepted on the transit network.
- Integrity / data-origin authentication — esp-sha-hmac is configured to detect any modification of packets in transit and to confirm they genuinely originated from the expected peer.
- Peer authentication — the pre-shared key configuration (Section 10), bound to specific peer IP addresses, is intended to ensure R1 and R3 only build a tunnel with each other, not with an unauthorized third party.
- Perfect Forward Secrecy — set pfs group24 on the crypto map is configured so that each new IPsec SA derives its own independent key material, limiting the impact of any future key compromise on past sessions.
- Segregation from transit infrastructure — R2 holds no cryptographic configuration and cannot decrypt the protected traffic it forwards; in a design where the tunnel is actually established, this means even a compromised transit router could not read LAN-to-LAN traffic.

These are the properties the configuration is designed to provide. As documented in Sections 14–17, we cannot claim from our own evidence that an active, functioning tunnel was actually protecting traffic at the time of testing; the security properties above should be read as the intended design outcome of a correctly matched configuration, not as an empirically confirmed result of this specific run.

20. Troubleshooting and Observations

Observed Issue: IKE/IPsec SA Not Established

1. Initial symptom: after applying the crypto map to R1's G0/0/0 interface and generating interesting traffic (PC-A → PC-C, matching ACL 101), show crypto isakmp sa continued to return an empty SA table, and show crypto ipsec sa continued to report zero packet counters and an unassigned outbound SPI, both before and after the test traffic (Figures 9 and 11).
2. Tests performed: (a) confirmed ISAKMP policy parameters matched on both R1 and R3 (Figure 6); (b) confirmed transform-set algorithm parameters matched on both routers, despite a naming inconsistency (Section 11); (c) confirmed ACL 101 was correctly mirrored on both peers (Figure 7); (d) confirmed the crypto map's peer, ACL, transform-set, PFS, and interface binding on R1 (Figure 8); (e) confirmed IP-layer connectivity existed independently of the VPN via a full PDU sweep (Figure 5).
3. Findings: every individually inspectable configuration element matched the lab's intended design and was consistent between R1 and R3, with two documented deviations from the lab's exact instructions — the ISAKMP Diffie-Hellman group reported as #1 (768-bit) rather than the instructed group 24 (Section 9), and a transform-set tag mismatch between R1 (50) and R3 (50 and R1-R3) that does not affect negotiated parameters (Section 11). Neither deviation, on its own, should categorically prevent SA formation, since the relevant negotiable values (encryption, hash, auth method, DH group for Phase 1; transform contents for Phase 2) were identical on both peers in our captures.
4. Corrective action: given the scope of this submission, no further corrective action (such as re-issuing debug crypto isakmp, clearing the crypto map and reapplying it, or re-verifying the pre-shared key character-for-character on both routers) is captured in the provided evidence. These are the logical next diagnostic steps and are noted here as unresolved rather than performed.

5. Final result: the underlying network (Parts 1) is fully functional and verified. The VPN configuration (Part 2) is complete and internally consistent, but this submission's evidence does not demonstrate an established Security Association or confirmed encrypted traffic; this is reported as an open item rather than a success, in keeping with the requirement not to claim results the evidence does not support.

21. Configuration Summary

Routing

- OSPF process 101, area 0, enabled on R1, R2, and R3, advertising 192.168.1.0/24, 10.1.1.0/30, 10.2.2.0/30, and 192.168.3.0/24 as appropriate per router.

IKE Phase 1 (ISAKMP Policy 10)

- Authentication: pre-shared key
- Encryption: AES 256
- Hash: SHA
- DH group: configured intent = group 24; verified output = group #1 (768-bit) — see Section 9
- Lifetime: 3600 seconds

IPsec Phase 2

- Transform set: esp-256-aes esp-sha-hmac, Tunnel mode (tagged 50 on R1; 50 and R1-R3 on R3)
- Encryption: AES-256 (ESP)
- Integrity: HMAC-SHA
- SA lifetime: 1800 seconds (global); 900 seconds (crypto map override)

Traffic Selection

- ACL 101 — mirrored on R1 (192.168.1.0/24 → 192.168.3.0/24) and R3 (192.168.3.0/24 → 192.168.1.0/24)

VPN Peers

- R1 (10.1.1.1) ↔ R3 (10.2.2.1), via transit router R2

Crypto Map

- Name: CMAP, sequence 10, type ipsec-isakmp
- Peer: 10.2.2.1 (on R1) / 10.1.1.1 (on R3)
- ACL: 101 | Transform set: R1-R3-equivalent (Section 11) | PFS: group24
- Applied interface: R1 GigabitEthernet0/0/0; R3 GigabitEthernet0/0/0 (WAN-facing interface toward the peer in both cases)

22. Final Results

Component	Status
Network connectivity (IP addressing)	Confirmed — interfaces up/up, correct addressing (Figures 2–3)

Component	Status
OSPF routing	Confirmed — FULL adjacency and correct routes (Figure 4)
End-to-end IP connectivity (PC-A ↔ PC-C)	Confirmed — all PDUs Successful (Figure 5)
ISAKMP / crypto map configuration	Confirmed — all parameters present and consistent (Figures 6, 8)
IKE Phase 1 negotiation (SA established)	Not confirmed — empty SA table (Figures 9–11)
IPsec Phase 2 negotiation / encrypted traffic	Not confirmed — zero packet counters (Figures 9, 11)

The lab was substantially, but not completely, verified as successful based on the evidence gathered by Team 3. Every prerequisite and every configuration element required for a working Site-to-Site IPsec VPN was correctly built and independently confirmed with show commands. What could not be confirmed, specifically, was the final and most important proof point of the entire exercise: an actively negotiated, encrypting IPsec tunnel between R1 and R3.

23. Conclusion

This lab walked through the complete engineering chain behind a Cisco IOS Site-to-Site IPsec VPN: a working OSPF-routed underlay first, then an ISAKMP (IKE Phase 1) policy to authenticate the two peers and build a secure control channel, an IPsec (Phase 2) transform set to define how data would actually be encrypted, an access list to define precisely which traffic deserves that protection, and a crypto map that binds peer, traffic selector, and transform set together and activates the policy on the correct interface.

Team 3 successfully implemented and verified every configuration stage of this chain: addressing, OSPF adjacency and routing, ISAKMP policy parameters, the IPsec transform set, the mirrored interesting-traffic ACL, and the crypto map's binding to peer, ACL, transform set, PFS, and interface. Full IP-layer connectivity between PC-A and PC-C was also demonstrated. However, our own show crypto isakmp sa and show crypto ipsec sa captures, taken both before and after generating traffic that matches the crypto ACL, show no established Security Association and no non-zero encryption/decryption counters.

This distinguishes clearly between what this lab taught and what this specific implementation ultimately proved: the relationship between routing, interesting traffic, IKE, IPsec, crypto maps, Security Associations, and encrypted traffic is well understood and was correctly built at the configuration level, but the tunnel itself did not come up as evidenced in this submission. Reporting that outcome honestly — rather than assuming success because the surrounding configuration was correct — is itself part of the discipline this lab is meant to teach.